DE–2809 **11**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, DECEMBER 2011.

First Semester

PRINCIPLES OF INFORMATION TECHNOLOGY

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) List out the elements of computer and explain in detail.

   (b) Discuss about developments in communication technology.

2. (a) Discuss about ethics of information technology.

   (b) Write notes on connectivity and interactivity of communication technology.

3. (a) Explain the types of software.

   (b) Write notes on the following :

      (i) Web browsers

      (ii) Desktop accessories

      (iii) Presentation graphics software.

4. (a) Discuss in detail about email and chat facilities of Internet.

   (b) Write notes on practical uses of communication and connectivity.

5. (a) Explain about data comm software and its special features.

   (b) Discuss in detail about the factors affecting data transmission.

6. (a) Write notes on cyber ethics.

   (b) Write notes on the following :

      (i) Magnetic tapes

      (ii) Hard disks

      (iii) Optical disks.

7. (a) Write down the features of DBMS.

   (b) Explain the five generations of programming languages.

8. (a) Discuss about the steps of programming languages.

   (b) Explain the security issues of information system.

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DE–2810 **12**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, DECEMBER 2011.

First Semester

DIGITAL COMPUTER ORGANISATION

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

$(5 \times 20 = 100)$

All questions carry equal marks.

1. (a) (i) What do you mean by 'Radix' of a number system? (2)

(ii) With an example discuss how will you convert a decimal number into binary. (3)

(iii) Find the hexadecimal and octal equivalent of the binary number 01101011100.1101011. (3)

(iv) Represent the decimal number 8620 in BCD and excess–3 code. (2)

(b) (i) What do you mean by complement of a number system? Compare between 1's and 2's complements. (5)

(ii) Write a short note on Alphanumeric codes. (5)

2. (a) (i) Compare Boolean algebra with arithmetic and ordinary algebra. (6)

(ii) State and prove De Morgan's Theorems. (4)

(b) What do you mean by Prime implicants? Determine the prime implicants of the function $F(w, x, y, z) = \sum(1,4,6,7,9,10,11,12,15)$ using Quine-Mckluskey method. (10)

3. (a) Design a combinational circuit whose input is a four-bit number and whose output is the 2's complement of the input number. (10)

(b) Design a BCD-to decimal decoder. (10)

4. (a) Discuss the steps involved in the design of an Arithmetic logic unit. (12)

(b) Design a full adder using NAND gates only. (8)

5. (a) Write a short note on Binary parallel adder. (10)

(b) Discuss addition and subtraction in a parallel arithmetic element. (10)

6. (a) Draw the logic diagram of master slave flip-flop and explain. (10)

(b) Write a short note on BCD counter. (10)

7. Discuss on the following :

(a) Linear select memory organization

(b) Magnetic disk memories. (10 + 10)

8. (a) What do you mean by priority interrupt? Discuss. (8)

(b) With a block diagram discuss on parallel peripheral interface. (12)

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DE–2811 **13**

DISTANCE EDUCATIO

M.C.A. (S) DEGREE EXAMINATION,

DECEMBER 2011.

First Semester

C AND DATA STRUCTURES

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions. (5 × 20 = 100)

All questions carry equal marks.

1. (a) (i) Discuss the structure of a C program. (3)

(ii) What are the key words in C? What restrictions apply to their use? (4)

(iii) What is a variable? How can variables be characterized? (3)

(b) (i) What is an operator? Describe the different types of operators that are included in C. (5)

(ii) Summarize the rules that apply to expressions. (5)

2. (a) (i) What are the commonly used input/output functions in C? Discuss. (5)

(ii) Summarize the syntactic rules associated with the while statement. (5)

(b) (i) What rules apply to nested loops? Discuss. (5)

(ii) Compare the use of the switch statement with the use of nested if-else statements. (5)

3. (a) (i) What is a function? State the advantages to the use of functions. (5)

(ii) With an example discuss how a value is passed to a function through arguments. (5)

(b) Discuss on the following:

(i) C preprocessor

(ii) Multi dimensional array. (5 + 5)

4. (a) (i) What is Macros? Discuss. (5)

(ii) Write a short note on library functions. (5)

(b) (i) What is an array? How are arrays processed in C? (4)

(ii) How can a list of strings be stored within a two-dimensional array? Discuss. (6)

5. (a) (i) Discuss on dynamic memory allocation. (5)

(ii) What is a pointer? What kinds of objects can be associated with pointer variable? (5)

(b) (i) How does a structure differ from an array? How is an array of structures initialized? (5)

(ii) Define a structure consisting of two floating point members called real and imaginary. Include the tag complex within the definition. (5)

6. (a) What is meant by opening a data file? Summarize the different file types that can be specified by the fopen function. Give examples. (10)

(b) Discuss on the following:

(i) Operations on a pointer

(ii) Passing structure to a function. (5 + 5)

7. (a) What is a data structure? Discuss. (6)

(b) What is a queue? How the structure of queue different from stack? List the different applications of stack. Discuss any one. (14)

8. (a) What is a linked listed? Discuss how the elements are deleted in a doubly linked list. (10)

(b) What is a binary tree? List the applications of trees. Discuss any one in detail. (10)

DE–2812 **14**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, DECEMBER 2011.

First Semester

OFFICE AUTOMATION

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal Marks.

($5 \times 20 = 100$)

1. (a) Discuss about working with multiple windows environment.

(b) Write the procedure to add and remove an application in windows operating system.

2. (a) Explain about table handling mechanism in MS office.

(b) Discuss about formatting of text in MS-Word.

3. (a) Write the procedure to perform mail merge in MS-Word.

(b) Explain the spell checking feature of MS-Word.

4. (a) Write notes on : Numeric, string, Date and time functions in MS-Excel.

(b) Explain the various properties of chart object in MS-Excel.

5. (a) Discuss about Absolute, Relative and mixed References in MS-Excel.

(b) Explain about sorting in MS-Excel database.

6. (a) Write notes on : Menus available in power point.

(b) Discuss about Formatting Bullets and Numbering in MS Power Point.

7. (a) How will you insert a table in power point slide?

(b) Write notes on MS-Access components.

8. (a) Discuss on query object in MS-Access.

(b) How will you combine Excel data and charts in word Documents?

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DE–2813 **21**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, DECEMBER 2011.

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions. (5 × 20 = 100)

All questions carry equal marks.

1. (a) (i) What are the basic concepts of object oriented programming? (5)

   (ii) List the applications of object oriented programming. (5)

   (b) (i) How are data and functions organized in an object-oriented program? (5)

   (ii) Compare C and C++. (5)

2. (a) (i) With suitable examples discuss on enumerated and derived data types. (6)

   (ii) What do you mean by type compatibility? Discuss with an example. (4)

   (b) (i) What is a manipulator? Discuss with an example. (4)

   (ii) Write a short note on memory management operators. (6)

3. (a) With examples discuss the different control structure of C++. (10)

   (b) (i) What is an iterative statement? Discuss. (5)

   (ii) Discuss on multidimensional arrays. (5)

4. (a) How does a C++ structure differ from a C++ class? Discuss. (8)

   (b) An election is contested by six candidates. The candidates are numbered 1 to 6 and the voting is done by marking the candidates number on the ballot paper. Write a C++ program to read the ballots and count the votes cast for each candidate using an array variable count. (12)

5. (a) What is argument passing? With examples discuss pass-by-value. (10)

   (b) (i) Why we have to use overloading concept of a functions? Discuss. (5)

   (ii) What is default arguments? Discuss with example. (5)

6. (a) Write a C++ program to read a matrix of size m × n from the key board and display the same on the screen using functions. (10)

   (b) (i) What is a virtual function? Discuss. (5)

   (ii) With an example discuss the use of inline functions. (5)

7. (a) (i) What is a destructor? Describe the importance of destructors. (5)

   (ii) List the rules for overloading operators. (5)

   (b) Discuss the following :

   (i) Polymorphism

   (ii) Hierarchical inheritance. (5 + 5)

8. (a) (i) List the stream classes for console operations. (5)

   (ii) What are the steps involved in using a file in a C++ program? Discuss. (5)

   (b) (i) Discuss error handling during file operations. (5)

   (ii) What do you mean by command-line arguments? Discuss.(5)

DE–2814 **22**

DISTANCE EDUCATIO

M.C.A. (S) DEGREE EXAMINATION, DECEMBER 2011.

Second Semester

SYSTEMS ANALYSIS AND DESIGN

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks. ($5 \times 20 = 100$)

1. (a) Explain different types of systems. (10)

(b) Discuss the characteristics of a system. (10)

2. (a) Explain the concept of MIS and DSS. (10)

(b) Discuss the Interpersonal of skills a system analyst. (10)

3. (a) Why is it difficult to determine user requirements? Illustrate. (12)

(b) What are data flow diagrams? How do they differ from structure charts? (8)

4. (a) Elaborate on the steps in feasibility analysis. What consideration are involved in feasibility analysis. (15)

(b) Write a note on Decision Table and Decision tree. (5)

5. (a) Discuss how the systems design goes through logical and physical design. (10)

(b) Discuss the significance of Input design for data entry. (10)

6. (a) Discuss on classification of forms. (10)

(b) Explain about Organization of file. (10)

7. (a) State and explain the levels of Quality Assurance. (10)

(b) Discuss the HIPO and IPO charts. (10)

8. (a) Describe about conversion methods. (12)

(b) Write a note on major phases in Hardware and software selection. (8)

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DE – 2815

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DISTANCE EDUCATI

M.C.A.(S) DEGREE EXAMINATION, DECEMBER 2011.

Second Semester

VISUAL PROGRAMMING – I

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) What is property window? List out the common properties of Visual Basic controls. (10)

   (b) With suitable examples, explain the event procedures of Visual Basic. (10)

2. (a) How do you add and remove a control in a form? Discuss any five Visual Basic controls. (10)

   (b) What are determinate loops and indeterminate loops? Explain each kind of loop with an example program. (10)

3. (a) Write a short note on statements in V.B. (10)

   (b) What is explicit declaration? Discuss with an example program. (10)

4. (a) Discuss the various string functions in VB with an example program. (10)

   (b) Discuss the flex grid control with an example program. (10)

5. (a) Illustrate static and dynamic arrays with an example program. (10)

   (b) Describe briefly on procedures and sub procedures in VB. (10)

6. (a) Write a short note on MDI forms. (10)

   (b) Briefly discuss about global procedures and variables. (10)

7. (a) Write a brief note on drive, directory and life list boxes. (10)

   (b) How do you create classes in VB? Discuss. (10)

8. (a) Write a short note on file system controls. (10)

   (b) How do you implement an animation in VB? Discuss with an example program. (10)

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DE–2816

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DISTANCE EDUCATION

M.C.A. (Semester) DEGREE EXAMINATION,

DECEMBER 2011.

Second Semester

COMPUTER GRAPHICS

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the DDA line drawing algorithm in detail with diagram. (10)

   (b) Describe the region filling techniques in detail. (10)

2. (a) Describe the video display generation in detail. (10)

   (b) Write in detail about the applications of computer graphics. (10)

3. (a) Explain the viewing transformations in detail in 2D. (10)

   (b) Discuss about the transformation principles and convex polygon clipping. (10)

4. (a) Explain the three dimensional transformation (Translation and Mirror Reflection) in detail with diagram. (10)

   (b) Describe the composite transformations in detail. (10)

5. (a) Explain the clipping against a finite view volume in detail. (10)

   (b) Write in detail about Cohen Sutherland algorithm. (10)

6. (a) Discuss about the construction of three dimensional view. (10)

   (b) Describe the z-buffer algorithm in detail. (10)

7. (a) Explain the Sutherland Hodgman algorithm in detail with diagram. (10)

   (b) Write in detail about line segment clipping and point clipping with diagrams. (10)

8. (a) Describe the user's model in user interface design in detail. (10)

   (b) Discuss about command language and information display. (10)

DE–2817 **31**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2011.

Third Semester

DISCRETE MATHEMATICS

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Show that the formula $P \vee \neg(P \wedge Q)$ is a tautology. (6)

   (b) Obtain the principal conjunctive normal form of $P \to ((P \to Q) \wedge \neg(\neg Q \vee \neg P))$. (7)

   (c) Obtain the principal disjunctive normal form of $(\neg P \to R) \wedge (Q \rightleftarrows R)$. (7)

2. (a) Show that the propositions $\neg(P \wedge Q)$ and $\neg P \vee \neg Q$ are equivalent. (4)

   (b) Show that from pt

   (i) $(\exists x)(F(x) \wedge S(x)) \to (y)(M(y) \to W(y))$

   (ii) $(\exists y)(M(y) \wedge \neg W(y))$ the conclusion $(x)(F(x) \to \neg S(x))$ follows. (8)

   (c) Prove the Demorgan's law. (8)

3. (a) Prove that $A - B = A \cap \sim B$. (4)

   (b) Prove that $(A \times B) \cap (A \times C) = A \times (B \cap C)$. (8)

   (c) Show whether the following relations are transitive, reflexive, symmetric

   (i) $R_1 = \{(1, 1)\}$

   (ii) $R_2 = \{(1,2),(2,2)\}$

   (iii) $R_3 = \{(1,2),(2,3),(1,3),(2,1)\}$. (8)

4. (a) Let R and S be the following relations on $A = \{1,2,3\}$ $R = \{(1,1),(1,2),(2,3),(3,1),(3,3)\}$,

   $S = \{(1,2),(1,3),(2,1),(3,3)\}$.

   Find $R \cap S$, $R \cup S$, R^c, $R \circ S$ and S^2. (8)

   (b) Define maximal compatibility block, the compatibility relation on a set $\{x_1, x_2, \ldots x_6\}$ be given by the matrix. (8)

x_2	1				
x_3	1	1			
x_4	0	0	1		
x_5	0	0	1	1	
x_6	1	0	1	0	1

   Draw the graph and find the maximal compatibility blocks of the relation.

   (c) Prove that if R is an equivalence relation on a set A, then R^{-1} is also an equivalence relation on A. (4)

5. (a) If $f:R\to R$ defined by $f(x)=x^2-2$ and $g:R\to R$ defined by $g(x)=x+4$ find $f\circ g$ and $g\circ f$. (8)

(b) Let $f:R\to R$ be defined by $f(x)=2x-3$. If f is 1-1 and onto then find f^{-1}. (4)

(c) Define characteristic function of a set and show that $A\cap(B\cup C)=(A\cap B)\cup(A\cup C)$. (8)

6. (a) Define semigroup and give example. (4)

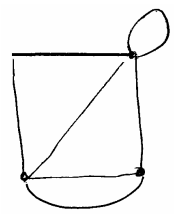
(b) State and prove Lagranges theorem. (8)

(c) A subset $S\neq\phi$ of G is a subgroup of $(G,*)$ iff for any pair of elements $a,b\in S$, $a*b^{-1}\in S$. (8)

7. (a) Define normal subgroup and give example. (4)

(b) Prove that the symmetric group S3 is not an abelian group. (8)

(c) Find incidence matrix and adjacency matrix for the following graph. (8)



8. (a) Define the terms

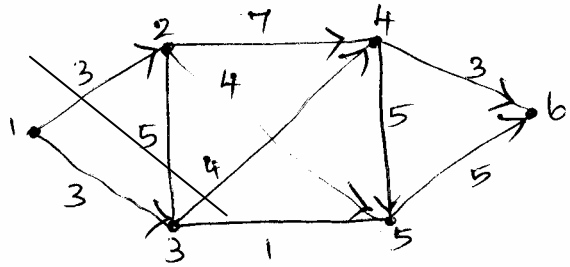
(i) Directed tree

(ii) Terminal tree

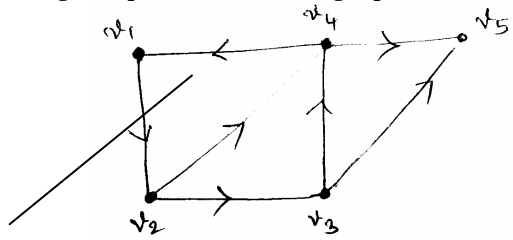
(iii) Ordered tree

(iv) Binary tree. (8)

(b) Find the shortest path for the following graph. (8)



(c) Find the strong components of the digraph. (4)



Also find its unilateral and weak components.

DE–2818

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DISTANCE EDUCATI

M.C.A.(S)/M.C.A.(Lateral) DEGREE EXAMINATION, DECEMBER. 2011.

Third Semester

OPERATING SYSTEMS

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) (i) What is an operating system? Discuss. (3)
   (ii) What is multiprogramming? What are the main advantages of multiprogramming? (4)
   (iii) Compare Batch and time-sharing operating systems. (3)
   (b) What is system calls? Discuss the different types of system calls. (10)

2. (a) What is a process? Discuss on process control Block. (10)
   (b) (i) What is a race condition? Discuss. (5)
   (ii) Write a short note on critical section problem. (5)

3. (a) (i) What is a schedules? Compare long term and short term scheduling. (5)
   (ii) What do you mean by preemptive and nonpreemptive scheduling? Compare. (5)
   (b) Compare any three process scheduling algorithms. (10)

4. (a) What is a device controller? Discuss. (10)
   (b) What is an interrupt? How you can classify interrupts? Write a short note on interrupt handlers. (10)

5. (a) (i) What is deadlock? Discuss the necessary conditions for deadlock. (4)
   (ii) Discuss on resource allocation graph. (6)
   (b) Write a short note on the following :
   (i) Buddy system
   (ii) Resident monitor. (5 + 5)

6. (a) What is paging? How page fault can be avoided? Discuss on paging hardware. (10)
   (b) (i) What is thrashing? Discuss. (4)
   (ii) Discuss multiprogramming with variable partitions. (6)

7. (a) (i) What is a file? What is the purpose of open and close operations. (5)
   (ii) Show how a sequential file is implemented on a direct access device like a disk. (5)
   (b) What is a directory? List the information which may be kept in a directory. Write a short note on tree-structured directories. (10)

8. (a) Write a short note on file protection. (10)
   (b) Compare the different disk scheduling algorithms. (10)

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DE–2819 **33**

DISTANCE EDUCATIO

M.C.A. (S)/ M.C.A. (Lateral) DEGREE EXAMINATION,

DECEMBER 2011.

Third Semester

RDBMS

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Notes on Schema. (10)

(b) Explain the features of DBMS. (10)

2. (a) Explain DDL operations. (10)

(b) Explain the concepts of relational model. (10)

3. (a) Explain the types of data bases. (10)

(b) Explain about the logical structures. (10)

4. (a) Notes on subqueries. (10)

(b) What are the types of expression in SQL? (10)

5. (a) What are the privileges required in SQL? (10)

(b) Notes on Embedded SQL statements. (10)

6. (a) What are the elements of SQL language? (10)

(b) Notes on physical files. (10)

7. (a) Explain the different user interfaces. (10)

(b) Notes on Relational model. (10)

8. (a) Notes on Indexes. (10)

(b) Explain about client/server systems. (10)

DE–2820 **34**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2011.

Third Semester

UNIX AND SHELL PROGRAMMING

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss about filters in UNIX. (10)

   (b) Write notes on directory level permissions. Explain about changing of permissions. (10)

2. (a) Discuss about the following : (10)

   (i) Date and time command

   (ii) ECHO command

   (iii) Uname command.

   (b) What is the command for each of the following tasks? Explain each command with proper example. (10)

   (i) List the contents of a directory

   (ii) Link a file to another

   (iii) Copy a file

   (iv) Display the absolute pathname of the current directory.

3. (a) Discuss about shell variables. (10)

   (b) Write notes on the following : (10)

   (i) Command arguments and parameters.

   (ii) Creation of new commands

   (iii) Redirecting to one file

4. (a) (i) Explain about conditional expression in shell programming. (10)

   (ii) Write notes on loop in shell programs.

   (b) Discuss about interrupts in shell programming. (10)

5. (a) Explain the following commands in shell programming :

   (i) Tap

   (ii) Pick

   (iii) News Command. (10)

   (b) Write about functions of command in shell programming. (10)

6. (a) (i) Explain about debugging in UNIX.

   (ii) Write notes on accessing the environment in UNIX. (10)

   (b) (i) Discuss about UNIX system calls.

   (ii) Describe about signals and interrupts in UNIX. (10)

7. (a) Discuss about Built-in functions.

   (b) Explain how functions and procedures are used in programming.

8. (a) Write notes on error recovery mechanisms in UNIX programming.

(b) Briefly discuss about document preparation tools.

9.

DE–2821 **35**

DISTANCE EDUCATION

M.C.A. (Semester)/M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2011.

Third Semester

COMMUNICATION SKILLS

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the various types and modes of communication in detail. (10)

   (b) Discuss about the concepts, definition and importance of communication. (10)

2. (a) Write in detail about the barriers in communication and commandments of communication. (10)

   (b) Discuss about the concept of minutes and communication aids. Also explain the improving English language writing in detail. (10)

3. (a) Discuss about the Art of listening and modes of expression and listening tests. (10)

   (b) Write in detail about methods of asking questions and art of conversation. (10)

4. (a) Explain the concept of brain storming and presentation reports in detail. (10)

   (b) Discuss about the types of interviews and effective communication during interviews. (10)

5. (a) Explain the art of guiding and controlling discussion in detail. (10)

   (b) Discuss about lateral thinking, participation techniques. (10)

6. (a) Explain the tool for personality identification and analysis of body language in detail. (10)

   (b) Discuss about desirable body language and body language as a powerful communication. (10)

7. (a) Describe the preparation and techniques for negotiation in detail. (10)

   (b) Discuss about the meaning, importance and objectives of meetings. (10)

8. (a) Write in detail about communication in management and communication within management. (10)

   (b) Describe the meaning and types of presentation in detail. (10)

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DE–2822 **41**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A.(Lateral) DEGREE EXAMINATION, DECEMBER 2011.

Fourth Semester

COMPUTER NETWORKS

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal Marks.

($5 \times 20 = 100$)

1. (a) Describe the TCP/IP Reference model. (12)

(b) Compare satellite communication with fiber optic communication. (8)

2. (a) Explain the architecture of ATM. (10)

(b) Explain how twisted pairs can be used as transmission media. (10)

3. (a) Discuss on error detection and correction. (12)

(b) Explain Petri Net models. (8)

4. (a) Bring out the concept of collision Free protocols. (10)

(b) Discuss on ICMP. (10)

5. Write a detailed essay on issues related with network layer design. (20)

6. (a) Discuss on routing in Ad Hoc Networks. (12)

(b) Write a note on Berkeley Sockets. (8)

7. (a) Explain about the concept of multiplexing. (10)

(b) Explain the header format of TCP. (10)

8. (a) Explain substitution and Transposition ciphers. (10)

(b) Write a note on Video data compression. (10)

DE– 2823 **42**

DISTANCE EDUCATION

M.C.A. (Semester) /M.C.A. (Lateral)

DEGREE EXAMINATION, DECEMBER 2011.

Fourth Semester

ACCOUNTING AND FINANCIAL MANAGEMENT

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks. (5 × 20 = 100)

1. From the figures stated below prepare a revenue account and a valuation balance sheet as at 31st December 1974 showing surplus for policy holders.

	Rs.
Life assurance fund (operating)	40,00,000
Premiums	25,00,000
Interest, dividends and rents	15,00,000
Consideration for annuities granted	1,00,000
Claims paid	3,00,000
Surplus on revaluation of reversion purchased	8,000
Bonus in reduction of premiums	5,000
Surrenders	1,00,000
Commission	50,000
Net liability on policies in force on 31.12.74	56,53,000

2. Alpha manufacturing company has drawn up the following profit and loss account for the year ended 31.3.2009

	Dr Rs.		Cr Rs.
To opening stock	26,000	By sales	1,60,000
To purchases	80,000	By closing stock	38,000
To wages	24,000		
To manufacturing expenses	16,000		
To gross profit c/d	52,000		
	1,98,000		1,98,000
To selling and dist expenses	4,000	by gross profit	52,000
To administrative expenses	22,800	by commission	
To general expenses	1,200	received	4,800
To value of furniture			
Loss by fire	800		
To net profit	28,000		
	56,800		56,800

You are required to find out :

(a) Gross profit ratio

(b) Net profit ratio

(c) Operating ratio

(d) Operating net profit to net sales ratio.

3. Explain the procedure for the preparation of cash flow statement.

4. From the following balances at hemshah general Insurance co. Ltd prepare the necessary revenue accounts and the profit and loss account in respect of the year 1971.

Rs. Rs.

Bad debts (fire)	5,000	Difference in exchange	300
Bad debits (marine)	10,000	profit on sale of land	60,000
Auditor's fees	2,000	fire premium less reinsurance	6,00,000
Director's fees	4,200	marine premiums less reinsurance	10,80,000
Share transfer fees	400		
Miscellaneous income	1,600	management expenses	
Fire fund (1-1-71)	2,50,000	fire	1,45,000
Marine fund (1-1-71)	8,20,000	marine	4,02,000
Claim's paid (fire)	1,40,000	claims outstanding on 1.1.71	
Claim's paid (marine)	3,00,000	fire	50,000
Commission paid (fire)	90,000	claims outstanding on 1.1.71	
Commission paid (marine)	1,08,000	(marine)	60,000
Additional reserve on 1.1.71 (fire)	50,000	commission earned on reinsurance ceded (fire)	30,000
Depreciation	35,000		
Interest, dividends etc received	19,000	marine	60,000

(a) The normal reserve required is 50% of not premium for fire and 100% of net premium for marine. In addition for fire 15% of the net premium is to be provided as additional reserve.

(b) The estimated liability in respect of outstanding claims due or intimated as on 31st December.

1971 was: fire Rs. 1,00,000, marine Rs. 1,40,000.

(c) The management expenses stated above are the direct expenses for the respective departments. In addition common expenses of Rs. 20,000 were incurred which must be charged to each of the departments in the ratio of premiums received.

(d) The following reinsurance premiums in respect of business accepted and ceded respectively have not been included in the above figures.

	Reinsurance accepted	Reinsurance coded
Fire	Rs. 25,000	Rs. 20,000
Marine	Rs. 60,000	Rs. 45,000

(e) The rate of commission in case of fire business is 15% of reinsurance premium accepted and in case of marine business it is 10% of reinsurance premium accepted.

5. (a) Explain the following italicized terms :

Financial statements reflect a combination of recorded facts, accounting conventions and personal Judgement.

(b) What is meant by constitute financial statements? State nature and limitations of each of the financial statements.

6. Make out a cash budget for October, November and December 1979 from the following information.

(a) Actual and budgeted purchases :

	Actual		Budget
July	40,000	October	50,000
August	50,000	November	40,000
September	45,000	December	35,000

(b) Actual and budgeted sales :

	Actual		Budget
July	90,000	October	90,000

August	80,000	November	80,000
September	85,000	December	80,000

(c) Actual and budgeted wages:

	Actual		Budgeted
July	25,000	October	30,000
August	30,000	November	30,000
September	35,000	December	35,000

(d) Actual and budgeted general expenses :

	Actual		Budgeted
July	7,000	October	8,000
August	6,000	November	6,000
September	5,000	December	7,000

(e) Incentive bonus payable in October, advance income tax in December. Plant to be purchased in November.

(f) 10% of purchases and sales are on cash basis.

(g) Rent Rs. 250 payable each month.

(h) Credit purchases are paid after one month and credit sales are collected after two months. Time lag in wages and general expenses half month.

(i) Cash and bank balances on 1st October 1979 – Rs. 40,000.

7. The board of directors of nanak engineering company private ltd request you to prepare a statement showing the working capital requirements forecast for a level of activity of 1,56,000 units of production.

The following information is available for your calculation :

(a)

	Per unit Rs.
Raw materials	90
Direct labour	40
Overheads	75
	205
	60
Profits	265

Selling price per unit :

(b) (i) Raw materials are in stock on average one month.
(ii) Materials are in process, an average 2 weeks.
(iii) Finished goods are in stock, on average 1 month.
(iv) Credit allowed by suppliers one month.
(v) Time lag in payment from debtors two months.
(vi) Log in payment of wages 1 ½ weeks
(vii) Log in payment of overheads is 1 months.

20% of the output is sold against cash. Cash in hand and at bank is expected to be Rs. 60,000. It is to be assumed that production is carried evenly throughout the year, wages and overheads accure similarly and a time period of 4 weeks is equivalent to a month.

8. From the information given below prepare flexible budget at 60% and 80% capacities :

	At 75% capacity Rs.
Variable overheads	

Indirect materials	7,500
Indirect labour	22,500
Semi - variable overheads:	
Electricity (40% fixed)	37,500
Repairs and maintenance (80% fixed)	3,750
Fixed overhead :	
Salaries	1,00,000
Insurance	5,000
Depreciation	25,000

DE–2824 **43**

DISTANCE EDUCATION

M.C.A. (Semester)/M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2011.

Fourth Semester

RESOURCE MANAGEMENT TECHNIQUES

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. Solve the following linear programming problems graphically

   Maximize $z = 45x_1 + 80x_2$

   Subject to the constraints :

   $5x_1 + 20x_2 \leq 400$

   $10x_1 + 15x_2 \leq 450$

   $x_1, x_2 \geq 0$. (20)

2. Solve the following assignment problem which minimizes the total man hours : (20)

		Men			
		A	B	C	D
	1	10	25	15	20
Jobs	2	15	30	5	15
	3	35	20	12	24
	4	17	25	24	10

3. The mean arrival rate to a service centre is 3 per hour. The mean service time is found to be 10 minutes per service. Assuming Poisson arrival and exponential service time, find

   (a) Utilisation factor for this service facility

   (b) Probability of two units in the system

   (c) Expected number of units in the queue

   (d) Expected time in minutes that a customer has to spend in the system. (20)

4. Give a brief summary of various types of queuing models with suitable examples. (20)

5. Arrivals at a station have been formed to follow Poisson process. The mean arrival rate is $m = 6$ units per hour. Simulate five hours of arrivals at the station. (20)

6. A project consists of a series of tasks labelled $A, B, \ldots H, I$ with the following relationships ($W < X, Y$ means X and Y cannot start until W is completed, $X, Y < W$ means W cannot start until both X and Y are completed). With this notation construct the network diagram having the following constraints : (20)

   $A < D, E;\ B, D < F;\ C < G;\ B < H;\ FG < I$.

   Find also the minimum time of completion of the project when the time (in days) of completion of each task is as follows :

Task :	A	B	C	D	E	F	G	H	I
Time :	23	8	20	16	24	18	19	4	10

7. A firm is considering replacement of an equipment, whose first cost is Rs. 1,750 and the scrap value is negligible at any year. Based on experience it was found that the maintenance cost is zero during the first year and it increases by Rs. 100 every year thereafter.
   (a) When should the equipment is replaced if annual interest rate = 0%
   (b) When should the equipment is replaced if annual interest rate = 12%.
   (20)
8. Explain the Resource Levelling by network techniques in detail, with examples. Also discuss about the two types of problems involved in resource levelling. (20)

DE–2825 **44**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2011.

Fourth Semester

OBJECT ORIENTED ANALYSIS AND DESIGN

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. What is system development methodology? How does object oriented approach differ from traditional approach in system development? (20)

2. Describe the various object oriented concepts. (20)

3. Explain the following :
   (a) Objects (4)
   (b) Classes (4)
   (c) Identity (4)
   (d) Object state and property (4)
   (e) Static binding and dynamic binding. (4)

4. For ATM example, show how actors and use case are identified? (20)

5. Give the importance of analysis and design and explain the view processes inception and elaboration. (20)

6. Draw a state diagram, activity diagram and package diagram for ViaNet bank ATM system. (20)

7. Describe the nature of an object and relationship among the objects. (20)

8. Draw the use case model for ATM operation. (20)

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DE–2826 **45**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2011.

Fourth Semester

INTERNET PROGRAMMING

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Explain about Microsoft Internet Implementation. (10)

   (b) Explain the WWW Design issues. (10)

2. (a) Explain the features of Scripting Languages. (10)

   (b) Write a note on Java Script. (10)

3. (a) Write in detail about SGML. (10)

   (b) What is HTML? Explain about HTML tags. (10)

4. Explain about ActiveX Control to activate a Web page. (20)

5. Explain about Netscape Navigator. (20)

6. (a) Explain the concept of VDO Live Technology. (10)

   (b) Discuss about CGI and PERL. (10)

7. Explain about pulling information through Web with an example. (20)

8. (a) Explain in detail about Images and Hyperlinks. (10)

   (b) Explain the uses of Animation. (10)

DE–2827 **51**

DISTANCE EDUCATI

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION,
DECEMBER 2011.
Fifth Semester
IMAGE PROCESSING AND ANALYSIS
(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.
All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Explain the elements of Digital geometry in detail. (10)

   (b) Describe the components of Image processing system in detail with diagrams. (10)

2. (a) Discuss about the Texture and pattern discrimination. (10)

   (b) Explain the Applications of digital image processing in detail. (10)

3. (a) Discuss about contrast Intensification and smoothing. (10)

   (b) Describe the mean filter and ordered statistic filter in detail. (10)

4. (a) Discuss about the Least-square Error Restoration. (10)

   (b) Write in detail about the Low-pass filtering and High-pass filtering. (10)

5. (a) Explain the block coding and quad tree coding in detail. (10)

   (b) Describe the plane-to-plane transformations mapping in detail. (10)

6. (a) Write in detail about medical image processing. (10)

   (b) Discuss about the Lossy compression methods and Loss less compression methods. (10)

7. (a) Explain the Edge elements extraction by Thresholding in detail. (10)

   (b) Describe the Threshold selection methods in detail. (10)

8. (a) Discuss about the clustering and statistical classification. (10)

   (b) Explain the fuzzy mathematical recognition in detail. (10)

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DE–2828 **52**

DISTANCE EDUCATION

M.C.A (S)/M.C.A (S) (Lateral) DEGREE EXAMINATION, DECEMBER 2011.

Fifth Semester

SOFTWARE ENGINEERING

(2005 to 2010 Calender Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Explain the software metrics in detail with examples. (10)

   (b) Describe the linear sequential model and rapid application development (RAD) model in detail with diagrams. (10)

2. (a) Write in detail about "spiral model" and 'prototyping' model with diagrams. (10)

   (b) Discuss about the role of management in software development. (10)

3. (a) Discuss about software project planning. (10)

   (b) Write in detail about software cost estimation model with examples. (10)

4. (a) Explain the software configuration management in detail. (10)

   (b) Describe the planning for quality assurance in detail. (10)

5. (a) Explain the information flow in detail with diagram. (10)

   (b) Describe the structured analysis in detail. (10)

6. (a) Write in detail about specification analysis. (10)

   (b) Discuss about structure specification techniques. (10)

7. (a) Explain the Architectural design in detail with diagrams. (10)

   (b) Discuss about transform mapping design with diagrams. (10)

8. (a) Describe the white box testing and basis path testing in detail with diagrams. (10)

   (b) Explain the black box testing and control structure testing in detail with diagrams. (10)

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DE–2829 **53**

DISTANCE EDUCATION

M.C.A (S)/M.C.A (S) (Lateral) DEGREE EXAMINATION, DECEMBER 2011.

Fifth Semester

VISUAL PROGRAMMING — II

(2005 to 2010 calender Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the files created by appwizard for windows program in detail with examples. (10)

   (b) Describe the mainframe object and hungarian notation in detail. (10)

2. (a) Discuss about the adding menus to the programs with examples. (10)

   (b) Write in detail about adding a caret to the window with examples. (10)

3. (a) Write in detail about connecting menu items to code with examples. (10)

   (b) Discuss about adding shortcut keys and adding Accelerator keys. (10)

4. (a) Explain the designing dialog boxes indetail with examples. (10)

   (b) Describe the creating a simple calculator application in VC++ in detail. (10)

5. (a) Discuss about list boxes and combo boxes with examples. (10)

   (b) Explain the creating a Notepad example in detail in VC++. (10)

6. (a) Discuss about reading files and writing files with examples. (10)

   (b) Write in detail about multiple views and serialization with examples. (10)

7. (a) Explain the debug windows in detail with examples. (10)

   (b) Write in detail about catching multiple exceptions with examples. (10)

8. (a) Describe the building ActiveX container application in detail with examples. (10)

   (b) Discuss about the break points and object linking and Embedding. (10)

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DE–2830 **54**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, DECEMBER 2011.

Fifth Semester

COMPILER DESIGN

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Discuss about the compilers and translators. (10)

(b) Explain the bookkeeping, error handling and compiler writing tools in detail. (10)

2. (a) Explain the role of the lexical analyzer in detail. (10)

(b) Discuss about the regular expression with examples. (10)

3. (a) Consider the ambiguous grammar

$S \to AS / b$

$A \to SA / a$

Construct the parsing table using SLR table construction algorithm. (10)

(b) Write the algorithm for constructing the SLR parsing table. (10)

4. (a) Consider the following grammar

$E \to E + T / T$

$T \to TF / F$

$F \to F * / a / b$ construct the LALR parsing table. (10)

(b) Explain the various steps involved in constructing canonical LR parsing tables in detail. (10)

5. (a) Discuss about quadruples and triples with examples. (10)

(b) Write in detail about postfix notations and postfix translations with examples. (10)

6. (a) Explain the representing scope information in detail with examples. (10)

(b) Describe the contents of a symbol table in detail. (10)

7. (a) Write in detail about lexical-phase errors with examples. (10)

(b) Discuss about the syntactic-phase errors with examples. (10)

8. (a) Write and explain the algorithm to construct a DAG in detail. (10)

(b) Explain the problems in code generation in detail. (10)

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DE–2831 **55**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, DECEMBER 2011.

Fifth Semester

DATA WAREHOUSING AND MINING

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks. ($5 \times 20 = 100$)

1. (a) Discuss about data mining and concept learning. (10)

(b) Write in detail about Information and production factor. (10)

2. (a) Explain the OLAP tools in detail. (10)

(b) Discuss about Neural networks. (10)

3. (a) Describe the Genetic algorithm in detail. (10)

(b) Describe the preliminary analysis of the data set using traditional query tools in detail. (10)

4. (a) Write in detail about data warehouse architecture with diagrams. (10)

(b) Discuss about process architecture. (10)

5. (a) Explain the partition strategy in detail. (10)

(b) Discuss about data marting and meta data. (10)

6. (a) Explain the hardware and operational design of data warehouses in detail. (10)

(b) Write in detail about physical layout of data warehouses. (10)

7. (a) Describe the operating the data warehouse in detail. (10)

(b) Discuss about the backup and recovery in data warehouses. (10)

8. (a) Explain the tuning the data warehouse in detail. (10)

(b) Describe the testing the data warehouses in detail. (10)

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DE–2832 **61**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, DECEMBER 2011.

Sixth Semester

SOFTWARE PROJECT MANAGEMENT

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the various types of projects in detail. (10)

(b) Describe the problems with software projects in detail. (10)

2. (a) Write in detail about Project Analysis and Technical Planning. (10)

(b) Discuss about the software estimation. (10)

3. (a) Explain the network planning models in detail. (10)

(b) Describe the shortening project duration in detail. (10)

4. (a) Discuss about sequencing and scheduling projects. (10)

(b) Write in detail about Activity planning. (10)

5. (a) Explain the risk monitoring and control in detail. (10)

(b) Discuss about planning for small projects. (10)

6. (a) Discuss about the basic functions of software configuration management. (10)

(b) Discuss about the responsibilities and standards in software configuration management. (10)

7. (a) Explain the prototyping in detail. (10)

(b) Describe the models in prototyping. (10)

8. Explain the case study of PRINCE project management in detail. (20)

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DE–2833 **62**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A.(S) (Lateral) DEGREE EXAMINATION, DECEMBER 2011.

Sixth Semester

MULTIMEDIA SYSTEMS

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the benefits and problems in multimedia in detail. (10)

   (b) Describe the cross platform compatibility in detail. (10)

2. (a) Write in detail about system components in multimedia. (10)

   (b) Discuss about the multimedia platforms. (10)

3. (a) Explain the analog, digital audio/video in detail. (10)

   (b) Discuss about the extended images and speech audio. (10)

4. (a) Write in detail about evaluating a compression system. (10)

   (b) Describe the JPEG and DVI in detail. (10)

5. (a) Explain the motion picture expert group in detail. (10)

   (b) Discuss about redundancy and visibility. (10)

6. (a) Explain the classes and related items in object oriented multimedia in detail. (10)

   (b) Discuss about the overview of multimedia frameworks. (10)

7. (a) Explain the media in real world applications in detail. (10)

   (b) Describe the training and education using multimedia in detail. (10)

8. (a) Write in detail about media organizaiton. (10)

   (b) Explain the media architecture and operations in detail. (10)

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DE–2834 **63**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, DECEMBER 2011.

Sixth Semester

NEURAL NETWORKS

(2005 onwards)

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Explain the benefits of Artificial Neural Networks in detail. (10)

(b) Discuss about the concept of knowledge representation. (10)

2. (a) Explain the construction of neural networks in detail with diagram. (10)

(b) Describe the biological neural networks in detail with diagram. (10)

3. (a) Explain the Boltzmann machine concept in detail. (10)

(b) Describe the Hebbian Learning in detail. (10)

4. (a) Discuss about the Learning process. (10)

(b) Explain the Supervised Learning in detail. (10)

5. (a) Discuss about Training Algorithms in back propagation neural networks. (10)

(b) Explain the applications of counter propagation networks in detail. (10)

6. (a) Describe the applications of Associative memory in detail. (10)

(b) Explain the Bidirectional Associative memory in detail with diagram. (10)

7. (a) Describe the optical neural networks in detail with diagram. (10)

(b) Discuss about the cognition concepts. (10)

8. (a) Write in detail about Adaptive Resonance Theory. (10)

(b) Discuss about Holographic correlators. (10)

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DE–2835 **64**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, DECEMBER 2011.

Sixth Semester

C SHARP (C#)

(2005 to 2010 calender year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the concept of code reuse in detail. (10)

   (b) Describe the concept of error handling in detail. (10)

2. (a) Write in detail about the evolution of C#. (10)

   (b) Discuss about the concept of type safety and security. (10)

3. (a) Explain the intermediate language and meta data in C# in detail. (10)

   (b) Describe the VOS type system in detail. (10)

4. (a) Write in detail about the common language specification in C#. (10)

   (b) Discuss about the concept of virtual execution system in C#. (10)

5. (a) Write in detail about C# data types with examples. (10)

   (b) Describe the concept of enumeration in C# in detail. (10)

6. (a) Explain the concept of events in C# in detail with example programs. (10)

   (b) Describe the destructors in C# in detail with example programs. (10)

7. (a) Write in detail about Do's and Don'ts of exception handling. (10)

   (b) Describe the control statements in C# in detail with example programs. (10)

8. (a) Discuss about documentation and comments in XML with examples. (10)

   (b) Explain the debugging and setting break points in C# in detail with example programs. (10)

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DE–3222 **15**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION,

DECEMBER 2011.

First Semester

LAB I — DATA STRUCTURES USING C

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

Examiner should select and give ONE question to each candidate by lot system.

1. (a) Write a menu driven program in C to find an element in an array using linear and binary search method.

(b) Write a C program to sort a set of elements using insertion sort.

Cut here

2. (a) Write a C pro[illegible] ment enqueue and dequeue operations on queue using linked list.

(b) Write a C program to evaluate the given mathematical expression using stack.

3. (a) Write a C program to implement string manipulation using switch case.

(b) Write a C program to find whether the given string is palindrome or not.

Cut here

4. (a) Write a C prog[illegible] ent push and POP operations on stack.

(b) Write a C program to find the factorial of a given number using recursion.

Cut here

5. (a) Write a C program to read 10 values to an assay variable. Use pointers to locate and display each value.

(b) Write a C program to copy contents of one file to another file.

Cut here

6. (a) Write a C program to implement insert and delete operations using linked list.

(b) Write a C program to sort 10 NOS is ascending order with naming of variable and the value before and after sorting.

Cut here

7. (a) Write a C progr[illegible] nt matrix manipulation.

(b) Write a C program to find the sum of digits in an array.

8. (a) Write a C program to find the roots of a quadratic equation $ax^2 + bx + c = 0$.

(b) Write a C program to find maximum and minimum among 'N' numbers.

DE–3223 **16**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, DECEMBER 2011.

First Semester

LAB II — MS OFFICE AND INTERNET

(2005 to upto 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Break-up of marks :

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

Answer any TWO questions choosing ONE from Section A and another from Section B.

Each question carries 50 marks.

SECTION A

1. Type a lecture note of [illegible] provide audio explanation with the help of sound files.

Cut here

2. Create a table of stude[illegible]ntains REG NO, NAME, SUBJECT1, SUBJECT2 and SUBJECT3 marks. Add a new column named TOTAL and find the total for 3 subjects of each student. Add a new column AVERAGE to find the average of 3 marks of each student. Add 2 rows named TOTAL, AVERAGE to find the total and average value for each subject. Convert the table to text.

Cut here

3. Create a form letter that informs the candidates about the date of interview for admission into your college. Mail merge it with an Access Database containing all the candidates information.

4. Create a work sheet wi[illegible] columns :

Salesman number, Salesma[illegible] product name, Sale amount.

Add three records for 5 different salesman who have carried out sales of different products in different cities. Find the following :

(a) City wise total sales.

(b) Salesman wise total sales.

(c) Product wise total sales.

(d) Grand total sales.

Cut here

5. Create a work sheet in [illegible] following columns :

Employee number, Name, [illegible]sic Pay, HRA, DA, LIC, PF, Gross Pay, Net Pay.

(a) Type data for empno, empname, designation, Basic pay, LIC and PF.

(b) Calculate

HRA = 20% of Basic Pay

DA = 25% of Basic Pay

Gross Pay = Basic Pay + HRA + DA

Net Pay = Gross Pay – (LIC + PF)

(c) Draw the bar chart between emp name and net pay.

(d) Sort the designation column and employee name column at a time.

6. Create STUDENT table with the following fields :

REG NO, NAME, MARK1, MARK2, MARK3.

Create ADDRESS table with fields REG NO, STREET, CITY and PIN.

(a) Write a query to display REG NO, NAME and total of all the 3 subject marks.

(b) Write a query to display REG NO, NAME, STREET, CITY, PIN and total of all the 3 subject marks.

Cut here

7. Create two tables SALE[illegible]d join them to produce a third table SALES3.

8. Create Inventory tab[illegible] ITEM NO, NAME, QUANTITY ON HAND, REORDER LEVEL.

(a) Create a form in custom format.

(b) Create a query REORDER to show all the items.

(c) Create a macro that executes REORDER query automatically.

Cut here

[illegible]CTION B

9. Create HTML pages to play Video and Audio files.

10. Create HTML page tha[illegible] reen space into three frames (one vertical and two horizontal of equal [illegible] three different files in them with scrolling titles relevant to their contents.

Cut here

11. Create HTML page to [illegible] es and their highlighted features, conducted by an educational institution in a colorful format.

Cut here

DE–3224 **25**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, DECEMBER 2011.

Second Semester

Lab III — GRAPHICS USING C++

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Break up of Marks:

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

Answer TWO questions one from Section A and another from Section B chosen by the candidates through lot system.

Write C++ programs and execute them.

SECTION A

1. Write an object oriented program to implement Bresenham's line drawing algorithm.

Cut here

2. Write a program to pe ensional geometric transformations over a closed shape through functions passed with necessary parameters.

3. Write a program to draw a polygon with arbitrary number of points.

4. Write a program to draw the shape of a car and perform translation and rotation of wheels through anim

Cut here

5. Draw a triangle shape d breadth values passed to a function.

Cut here

Cut here

SECTION B

6. Write a program to draw a closed shape and apply rotation and scaling transformations in forward and reverse directions to restore original shape and position.

7. Write a program to imp clipping algorithm.

Cut here

8. Write a program to draw different geometric shapes using virtual function.

9. Write a program that demonstrates scaling, translation and rotation of a triangle shape.

10. Write a program to draw non-linear objects using arcs.

Cut here

Cut here

DE–3225 **26**

DISTANCE EDUCATION

M.C.A (S) DEGREE EXAMINATION, DECEMBER 2011.

Second Semester

LAB IV — VISUAL BASIC

(2005 to upto 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Marks break up :

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

Examiner should select and give ONE question to each candidate by lot system.

Each subdivision carries 50 marks.

1. (a) Write a VB project that stores 10 student records with fields STUNO, STUNAME, AGE, SEX and consolidated percentage of marks, in an array. Display data fields in text boxes and provide command buttons to move to desired record.

(b) Write a VB project using control array that creates an ordinary calculator with appropriate command buttons. Include the following capabilities for the calculator : +, –, *, /, % .

2. (a) Write a VB project to create a screen saver that displays a list of pictures with 1 second pause in between successive pictures.

(b) Write a VB project for commercial bank operations using SB account database with the following features

(i) DEPOSIT AMOUNT

(ii) WITH DRAW AMOUNT

(with minimum balance condition)

Cut here

3. (a) Design and imp ive X control for stop watch used for sports.

(b) Write a VB project to display the availability of seats when the user selects the train name and date from the Combo box and timings from another Combo box.

Cut here

4. (a) Design and im tive X control for Railway clock (Hour display 1 to 24)

(b) Write a VB project to display the availability of seats when the user selects the Bus name and number from the Combo box and date and timings from another Combo box.

Cut here

5. (a) Create a VB ap ke the text bold, italic underline and also to color the text. (Using check box and option button).

(b) Write a VB project to display a message (Example left/right mouse button is clicked) When the mouse button is clicked using mouse move event of the form.

6. (a) Write a VB project to display the availability of books when the user selects the author name from the Combo box and book title from another Combo box.

(b) Design and implement one Active X control for monthly calendar.

Cut here

7. (a) Write a VB project using AX DLL or EXE add a class module that would perform the following functions :

(i) Sum of digits

(ii) Whether the given number is amstrong or not.

(b) Create an application using VB to move elements from one list to another list and add new items.

Cut here

8. (a) Write aVB proj bank that maintains a list of donors with address and their blood group. Provide the following reporting features.

Display all donors (using data reports)

(i) In age group 20-30

(ii) in a particular city

(b) Write a VB project to calculate the amount when the user selects the class 'type of the train' from the Combo box and also enters the number of passengers.

9. (a) Write a VB project to display the availability of seats when the user selects the flight name from the Combo box and date including timings from another Combo box.

(b) Design and implement one ActiveX control for digital electric current usage measurement meter used in houses.

Cut here

10. (a) Write a VB pro bank that maintains a list of donors with address and their blood group. Provide the following reporting features.

Display all the donors (using data reports)

(i) With particular blood group

(ii) Male donors with particular blood group

(iii) Female donors with particular blood group.

(b) Write a VB project for commercial bank operations using SB account database with the following features.

(i) ADD NEW ACCOUNT

(ii) Calculate simple interest and update balance taking average of last 6 month balance in the account

(iii) CLOSE ACCOUNT.

DE–3226 **36**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) Lateral DEGREE EXAMINATION, DECEMBER 2011.

Third Semester

LAB V — RDBMS

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Students has to answer one question by lot system.

Break up of Marks :

Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

1. A Daily Sales File contains records with the following :

Fields : Department number, date, item description, sale price for each item, quantity, cost of each item.

Write a program using oracle to list all the input data and compute total amount of sales and profit for each sale. Accumulate and print total of sales and profit. Show the output report in the following format :

Department number
Item description
Sales price
Quantity
Cost
Profit.

2. A hospital maintains blood donors record information in a file. The file contains the following fields :

Donor number, Donor name, Donor age, Donor address, Pincode, Place of birth, Blood group (A, B, AB and O). Write a program using oracle to print out the number, name and address of the donors of the following categories

(a) Blood donors having AB blood group.

(b) Blood donors in the age group between 16 to 25.

Cut here

3. Create a data base with as show below :

Student

REGNO, NAME, CITY, MARK

(a) Write query to increase the marks by 5 for all the students whose mark will not exceed hundred marks after this increment.

(b) Write query to order the data in descending order of marks.

(c) Write query to display all the data for students from a particular city.

Cut here

4. Create a table for savi nt to customers with suitable fields write PL/SQL program to perform money deposit and with drawal operations.

Write a PL/SQL program to prepare a mark statement of a student using Student Table and Mark Table. Assume your own data.

Cut here

5. A company RX Pvt. L thly salary to its employee. It consists of Basic Salary, Allowance, deduction D.A. – 43% of Basic Salary, H.R.A. – 7% of (Basic Salary + D.A.) Deductions : PF – subscribed by an capital LIC Premium – Payable by

employee, the salary saving scheme Loan Recovery- If any payable by an employee. Create a master table of 10 records with following fields : Employee number, Name, Designation, Basic Salary, HRA, DA, Bank A/C Number, LIC Premium No.

A transaction file contains :

Employee Number, P.F. Subcription, LIC Premium Amount, Loan Recovers.

Write a program in SQL to prepare a report with the following information : Serial number, Employee number, Bank A/C Number, Name Basic Salary, Total Allowances, Gross Pay, Total Deduction, Net Pay.

Cut here

6. Create a salesman tabl ving fields :

SMAN
SM Number
Name
Sale Amount
Product Code
Date of Sale

Write PL/SQL program to find the following :

(a) Total Sales Salesman Wise
(b) Total Sales Product Wise.

7. Create three tables in oracle with the following fields :

Cricket	Football	Hockey
Player No.	Player No.	Player No.
Name	Name	Name
Age	Age	Age
Number of Matches	Number of Matches	Number of Matches
Runs	Goals	Goals

(a) Write query to display the players playing all the three games.
(b) Write query to display the cross product of all the tables.

Cut here

8. Create a sales man tabl wing fields :

SMAN
SM Number
Name
Sale Amount
Product Code
Date of Sale

Write PL/SQL program to find the following :

(a) Total Sales of all Salesman in January.
(b) Total Sales.
(c) Total Sales of all Products in February.

Cut here

~~9. Write a PL/SQL progra~~ ~~monthly attendance statem~~ent of a student using student Table and Attendance table. Assume your own data.

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2011.

Third Semester

LAB VI — ALGORITHMS AND SHELL PROGRAMMING

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Breakup of marks :

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

ONE question is to be given by the examiner to each candidate by lot system.

1. (a) Write program in C/C++ for travelling salesman problem using branch and bound method.

(b) A hospital maintains patient details. Write shell program to list the following.

(i) Patients in a blood group wise.

(ii) Patients in age between 50 and 60

(iii) Exit.

2. (a) Write a shell program to prepare electric bill for domestic customers.

For first 100 units - Rs. 0.75/unit

For next 100 units - Rs. 1.50/unit

Above 200 units - Rs. 3.00/unit

Prepare the electricity bill in the following format.

Customer No : ————————

Customer Name : ————————

Previous reading : ————————

Current reading : ————————

Units consumed : ————————

Charge : ————————

(b) Write a C/C++ program to solve towers of Hanoi problem.

Cut here

3. (a) Write a C/C++ aight selection sorting.

(b) Write a shell program which accepts the name of a file from the standard input and perform the following test :

(i) Enter 8 names in a file.

(ii) Sort the names present in the file.

(iii) List unsorted and sorted file.

(iv) Quit.

4. (a) Write a shell program which periodically monitors the disk for the existence of a file and then execute the program once the file has been located. (Use UNTIL statement).

(b) Write a program for quick sort using divide and conquer algorithm. Test the program with sample data.

Cut here

5. (a) Write a C/C++ program to solve 'knight tour problem using back tracking method and execute the program.

(b) Write a menu driven shell program for the following :

(i) List of files.

(ii) Processes of users.

(iii) Today's date.

(iv) Users of system.

(v) Quit to UNIX.

Cut here

6. (a) Write a C/C++ ear and binary search and execute the program.

(b) Write a menu driven shell program to perform the following tasks.

(i) Enter the sentence in a file.

(ii) Search a given whole word in an existing file.

(iii) Quit.

7. (a) Write a shell program using 3 arguments to take the pattern as well as input and output file names. If the pattern is found, display the message "pattern found" else display "pattern not found". Also check the right number of arguments and print the error message if no. of arguments is not equal to three.

(b) Write a C/C++ program for straight insertion sorting.

Cut here

8. (a) Write a C/C++ it fibonacci series.

(b) Write and execute a shell program to do the following :

(i) Convert given string in center of given row in bold, normal or reverse video.

(ii) Write given string at given row column bold, normal or reverse video.

(iii) Search a given string using awk with various options.

Cut here

9. (a) Write and exec am to print monthly payslips for employees in an organization. Test the program with sample data and print the results.

(b) Write a C/C++ program for Heap sort.

Cut here

10. (a) Write C/C++ pr ght exchange sorting.

(b) Write a shell program for four function calculator by using y acc. command.

DE–3228 **46**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2011.

Fourth Semester

LAB VII — INTERNET PROGRAMMING

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Breakup of marks

Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

One question is to be given by the examiner to each candidate by lot system.

Each subdivision carries 50 marks.

1. (a) Write a Java program to create a color as the background.

(b) Create a web page in the format of front page of a newspaper using text links/ Align the text with colors.

Cut here

2. (a) Design and pul[illegible] for a college.

(b) Develop a picture gallery having atleast 5 pages. Each of them is having several pictures with suitable information.

3. (a) Write a swing program to create Buttons with
   (i) Tool tip next
   (ii) Image
   (iii) Border
   (iv) Short cut key.

(b) Create a screen with a string "web design by using atleast 20 HTML tags.

Cut here

4. (a) Write a VB scr[illegible] owing
   (i) Check the given passwora
   (ii) Change the existing password.

(b) Write a Java script to create a window by using the confirm message.

Cut here

5. (a) Write a Java s[illegible] rder form to select the house articles.

(b) Write a menu driver Java program to do database functions using database connectivity facility. Include display, edit, insert and delete. Functions from a table.

Cut here

6. (a) Write a VB scr[illegible] alculator for given month and year.

(b) Write a swing program to create the tabbed panels.

7. (a) Write a VB script to do the following:
   (i) Display the current date
   (ii) Find the difference between two dates
   (iii) Find the age of a person by providing data of birth.
   (Use dd/mm/yy format for date)

(b) Develop a single page advertisement for a shop to be opened shortly.

Cut here

8. (a) Develop a web [illegible] ruitment agency in an IT industry.

(b) Write a JavaScript to create a color palet and display the text in the color chosen from the palet with proper background color.

9. (a) Write a Java program using Applet to display any 3 images when 3 buttons in the Border layout are clicked. The image should be displayed at the center.

Cut here

(b) Write a Java program using Applet to display the different color and fonts.

10. (a) Write a Java program using Applet to display the dialogue menu in Applet.

Cut here

(b) Write a Java program using Applet to create the frames and its controls.

DE-3229 **47**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, DECEMBER 2011

Fourth Semester

LAB VIII — NETWORK LAB

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any ONE question chosen by the examiner through lot system.

Break-Up of Marks

Record Note Book	:	5
Algorithm/Flowchart	:	15
Program	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

1. (a) Write a Java program to add two numbers using RMI client, server technique.
   (b) Write a Java Program to find factorial of given number using RMI client, server technique.

2. (a) Write a Java Program to find factorial of given number using FACTORIAL INTERFACE Technique.
   (b) Write a Java Program to setup a chat server and chart client.

3. ---------- Cut here ----------

4. (a) Write a Java Program to display the users and their IP addresses logged in the network.
   (b) Write a Java program to setup an echo server and echo client.

---------- Cut here ----------

5. (a) Write a Java P RYPT given word.
   (b) Write a Java Program to setup Time Server and Time Client.

---------- Cut here ----------

6. (a) Write a Java Program to send Message from one system to another system.
   (b) Write a Java Program to read a file from the remote system.

---------- Cut here ----------

7. (a) Write a Java Program to Subtract two numbers using RMI Client, server technique.
   (b) Write a Java Program to multiply two numbers using RMi client, server technique.

8. (a) Write a Java P Cut here e two numbers using RMI client, server technique.
   (b) Write a Java Program to decrypt the given encrypted word.

9. (a) Write a Java Program to find the given number is Armstrong or not using RMI client, server technique.
   (b) Write a Java Program to find the given word is palindrome or not using RMI client, server technique.

---------- Cut here ----------

10. (a) Write a Java P e the Towers of Hanoi problem using RMI client, server technique.

(b) Write a Java Program to send messages from one system to a group of systems in the LAN network.

Cut here

11. (a) Write a Java P the square of a number using RMI client, server technique.

(b) Write a Java Program to find the given number is palindrome or not using RMI client, server technique.

DE–3230 **56**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION,

DECEMBER 2011.

Fifth Semester

VISUAL C++ – LAB IX

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

Break-Up of Marks

Record Note Book	:	5
Algorithm/Flowchart	:	15
Program	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

1. (a) Write Visual C++ win 32 application program using MFC that displays lines, rectangle, triangle to draw a house.

   (b) Write Visual C++ win 32 application program using MFC that displays the message "GOOD MORNING" wherever the user clicks the mouse button on the client area.

2.

3. (a) Write visual C++ win 32 application program using MFC that displays carets of different shapes on direct area.

   (b) Write a visual C++ win 32 application program using MFC that creates a list box and displays names of the subject in the V semester.

---------- Cut here ----------

4. (a) Write a visual C++ win 32 application program using MFC that creates two buttons ok and CANCEL on the client area. Buttons should respond to use click over them and display appropriate message.

   (b) Write visual C++ program win 32 application using MFC that create a new font and display colors.

———————

DE-3231 **57**

DISTANCE EDUCATION

M.C.A. (S)/ M.C.A. (S) (Lateral) DEGREE EXAMINATION, DECEMBER 2011.

Fifth Semester

LAB X — COMPILER DESIGN

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

Break-Up of Marks

Record Note Book	:	5
Algorithm/Flowchart	:	15
Program	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

1. Write a C program to translated the expression $(a+b)*(b-c)/(a+b)$ into postfix notation, three-address code and quadruples.

2. Write a C program to translate the expression $(a+b)*(b-c)/(a+b)$ into infix expression and triples and indirect triples.

---------- Cut here ----------

3. Write a C program for the following .
   (a) Remove spaces
   (b) Comment lines
   (c) New lines and
   (d) Produce line numbers.

---------- Cut here ----------

4. Write a C program to accept a string and check whether if is an operator, grammar or not
   (a) $S \to aBCD;\ S \to aBdc$
   (b) $E \to E + E / E * E /(E)/ id$
   (c) $E \to EOP\ \ EE \to (E);\ E \to id;\ \ OP \to + \mid - \mid * \mid / .$

---------- Cut here ----------

5. Construct a C program to accept a string and check whether it satisfies the following production rule
   (By using operator precedence Grammar)
   $S \to AbB$
   $A \to aBb$
   $B \to a / Bba$.

6. Write a C program to accept a string and check whether it belongs to ambiguous grammar or not by using following production rule.
   E → E + E/E – E/E * E/E/E/id/const/(E) .

---------- Cut here ----------

7. Write a C program to construct the left-most derivation and right-most derivation for the following grammar.

$S \rightarrow a / \wedge / (T)$

$T \rightarrow T, S / S$

(a) $(a,(a,a))$

(b) To find the LEADING and TRAILING sets.

---------- Cut here ----------

8. Write a C program to construct predictive parsing table for the following production rule

S→iCtSS1/a

$S1 \rightarrow eS / E$ (E is empty)

$C \rightarrow b$.

Cut here

9. Write a C program to co LR parsing table for the following grammar

$E \rightarrow E + T / T$

$T \rightarrow T * F / F$

$F \rightarrow (E) / id$.

10. Write a C program to construct of an canonical LR parsing table for the following grammar.

$S \rightarrow CC$

$C \rightarrow CC / d$